IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

La era del acceso comercial al espacio ya es una realidad. Varias empresas están impulsando una nueva industria en la que los viajes y servicios espaciales son cada vez más accesibles y frecuentes.

Virgin Galactic ofrece vuelos suborbitales para turistas espaciales.

Rocket Lab se especializa en el lanzamiento de pequeños satélites.

Blue Origin desarrolla cohetes reutilizables con capacidad suborbital y orbital.

Entre todas ellas, SpaceX destaca como la más influyente, con hitos como el envío de naves a la Estación Espacial Internacional, el despliegue de la red de satélites Starlink para acceso global a internet y el transporte de astronautas en misiones tripuladas.

Uno de los factores clave del éxito de SpaceX es la reducción del coste de lanzamiento. Mientras otros proveedores superan habitualmente los 165 millones de dólares por misión, SpaceX anuncia lanzamientos del Falcon 9 por aproximadamente 62 millones de dólares. Gran parte de esta ventaja se debe a la reutilización de la primera etapa del cohete.

La primera etapa es el componente más grande y costoso del sistema. Realiza la mayor parte del trabajo durante el despegue y puede regresar a la Tierra para ser reutilizada en futuras misiones. Sin embargo, no todos los aterrizajes son exitosos, ya que dependen de las condiciones de la misión, la carga útil o los requisitos del cliente.

Para comprender mejor su estructura y escala, se utilizan las infografías de Forest Katsch, diseñador 3D e ingeniero de software que crea visualizaciones sobre cohetería y exploración espacial.

Introduction

En este proyecto final asumiré el rol de científico de datos en una empresa ficticia llamada Space Y, creada con el objetivo de competir con SpaceX.

El objetivo principal es estimar el coste de cada lanzamiento y comprender los factores que influyen en su precio. Para ello, trabajaré con datos públicos y desarrollaré paneles de control que ayuden a analizar el comportamiento de los lanzamientos.

Además, uno de los retos clave será predecir si la primera etapa del cohete podrá aterrizar y ser reutilizada con éxito. Esta predicción se abordará mediante técnicas de aprendizaje automático, en lugar de métodos tradicionales de ingeniería aeroespacial.

En resumen, los problemas principales a resolver son:

- Analizar los factores que determinan el coste de un lanzamiento espacial
- Estimar la probabilidad de éxito en el aterrizaje de la primera etapa
- Evaluar el potencial de reutilización del cohete como factor de reducción de costes
- Construir modelos predictivos basados en datos históricos de SpaceX
- Generar herramientas de visualización para apoyar la toma de decisiones del equipo

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- En este proyecto se utilizan datos de lanzamientos de SpaceX obtenidos mediante la API REST de SpaceX. El objetivo es analizar la información de los lanzamientos para predecir si un cohete intentará aterrizar.
- Los datos se obtienen en formato JSON y se transforman en un DataFrame para su análisis. Además, se complementan con técnicas de web scraping usando BeautifulSoup.
- Finalmente, se limpian y preparan los datos eliminando lanzamientos no deseados, gestionando valores nulos y organizando la información para su uso en modelos predictivos.

Data Collection – SpaceX API

- Request and parse the SpaceX launch data using the GET request.
- Filter the dataframe to only include Falcon 9 launches.

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
1	2010-06-04	Falcon 9	6123.547647058824	LEO	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B0003	-80.577366	28.5618571
2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B0005	-80.577366	28.5618571
3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B0007	-80.577366	28.5618571
4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False		1.0	0	B1003	-120.610829	34.632093
5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B1004	-80.577366	28.5618571
6	2014-01-06	Falcon 9	3325.0	GTO	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B1005	-80.577366	28.5618571
7	2014-04-18	Falcon 9	2296.0	ISS	CCSFS SLC 40	True Ocean	1	False	False	True		1.0	0	B1006	-80.577366	28.5618571

Place your flowchart of SpaceX API calls here

API SpaceX (launches/past)

↓

Solicitud GET (requests)

↓

Respuesta JSON (lista de lanzamientos)

↓

json_normalize → DataFrame

↓

Enriquecimiento de datos

(API adicional: cores, payload, booster, launchpad)

Data Collection - Scraping

- Web scraping Falcon 9 and Falcon Heavy Launches.
- Records from Wikipedia
  Request the Falcon9 Launch Wiki page from its URL.
- Extract all column/variable names from the HTML table header
- Create a data frame by parsing the launch HTML tables.

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
1	2010-06-04	Falcon 9	6123.547647058824	LEO	CCSFS SLC 40	None None	1	False	False	False		1.0	0	B0003	-80.577366	28.5618571
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7	2014-04-18	Falcon 9	2296.0	ISS	CCSFS SLC 40	True Ocean	1	False	False	True		1.0	0	B1006	-80.577366	28.5618571

Place your flowchart of web scraping here

Web Scraping (BeautifulSoup)

↓

Limpieza de datos

- Filtrar Falcon 9

- Eliminar Falcon 1

- Tratar valores nulos (PayloadMass)

↓

Dataset final limpio

↓

Análisis + predicción de aterrizaje

Data Wrangling

Using mainly Pandas' value_counts method, we calculated the following:

- Describe how data were processed
- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurence of mission outcome per orbit type
- Create a landing outcome label from Outcome column

EDA with Data Visualization

- We can plot out the FlightNumber vs. PayloadMassand overlay the outcome of the launch.
- Visualize the relationship between Flight Number and Launch Site.
- Visualize the relationship between Payload and Launch Site.
- Visualize the relationship between success rate of each orbit type.
- Visualize the relationship between FlightNumber and Orbit type.
- Visualize the relationship between Payload and Orbit type.
- Visualize the launch success yearly trend

EDA with SQL

- Display the names of the unique launch sites in the space mission.
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was acheived.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster_versions which have carried the maximum payload mass. Use a subquery.
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
- Rank the count of successful landing_outcomes between the date 04- 06-2010 and 20-03-2017 in descending order.

Build an Interactive Map with Folium

In this exploratory data analysis labs, I have visualized the SpaceX launch dataset using matplotlib and seaborn and discovered some preliminary correlations between the launch site and success rates. In this lab, I will be performing more interactive visual analytics using Folium:

- Mark all launch sites on a map.
- Mark the success/failed launches for each site on the map.
- Calculate the distances between a launch site to its proximities.

Build a Dashboard with Plotly Dash

- Add a Launch Site Drop-down Input Component.
- Add a callback function to render success-pie-chart based on selected site dropdown.
- Add a Range Slider to Select Payload.
- Add a callback function to render the success-payload-scatter-chart scatter plot.

We conducted this lab to analyze and answer the following:

- Which site has the largest successful launches? KSC LC-39A (10 launches)
- Which site has the highest launch success rate? KSC LC-39A (76,90%)
- Which payload range(s) has the highest launch success rate? From 2K to 4K
- Which payload range(s) has the lowest launch success rate? From 9K to 10K
- Which F9 Booster version (v1.0, v1.1, FT, B4, B5, etc.) has the highest launch success rate? FT

Predictive Analysis (Classification)

Perform exploratory Data Analysis and determine Training Labels

- create a column for the class
- Standardize the data
- Split into training data and test data

Find best Hyperparameter for SVM, Classification Trees and Logistic Regression

- Find the method performs best using test data

•

Results

MODEL	BEST PARAMETERS	ACCURACY
Logistic Regression	tuned hpyerparameters :(best parameters) {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}	0.8333333333333334
Support Vector Machine	tuned hpyerparameters :(best parameters) {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}	0.8333333333333334
Decision Tree Classifier	tuned hpyerparameters :(best parameters) {'criterion': 'gini', 'max_depth': 8, 'max_features': 'sqrt', 'min_samples_leaf': 4, 'min_samples_split': 2, 'splitter': 'best'}	0.9444444444444444
K Nearest Neighbors	tuned hpyerparameters :(best parameters) {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}	0.8333333333333334

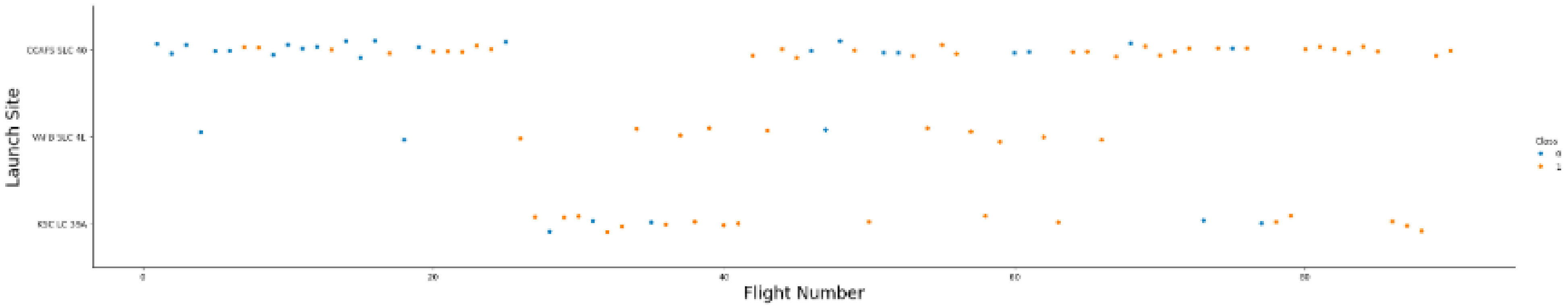
The method performs best is Decision Tree Classifier
with Accuracy: 0.9444444444444444 and False Positive = 1

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

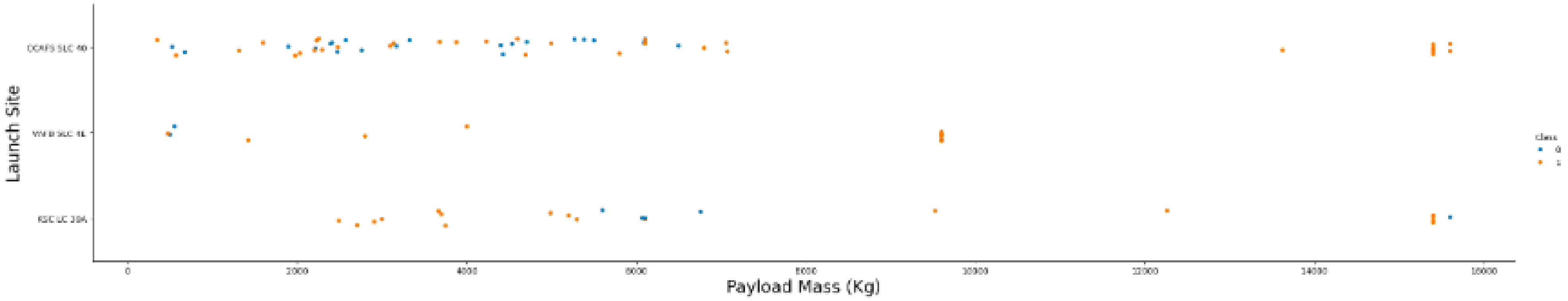
```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be t
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Launch Site",fontsize=20)
plt.show()
```



We can observe successful and unsuccessful launches based on the launch location. The largest number of pitches are at base CCAFS SLC40.

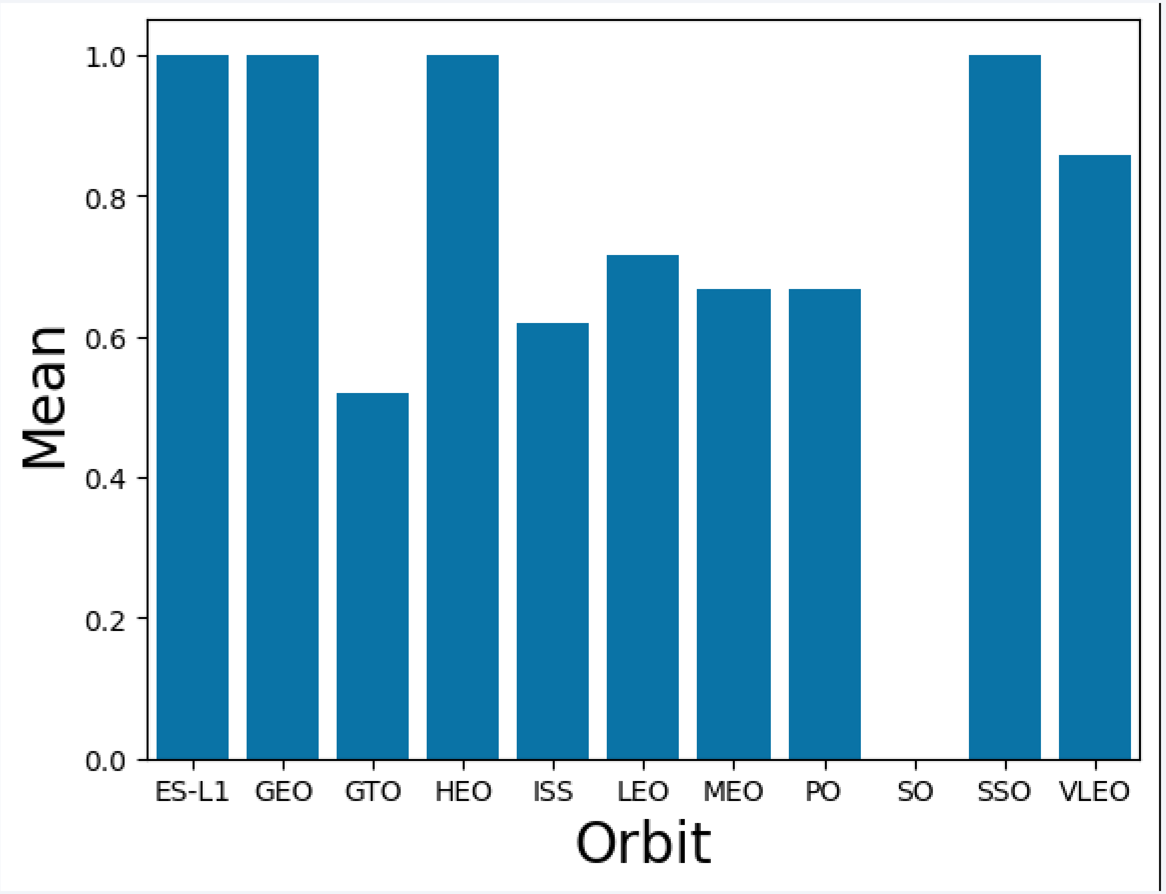
Payload vs. Launch Site

```python
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect = 5)
plt.xlabel("Payload Mass (Kg)",fontsize=20)
plt.ylabel("Launch Site",fontsize=20)
plt.show()
```



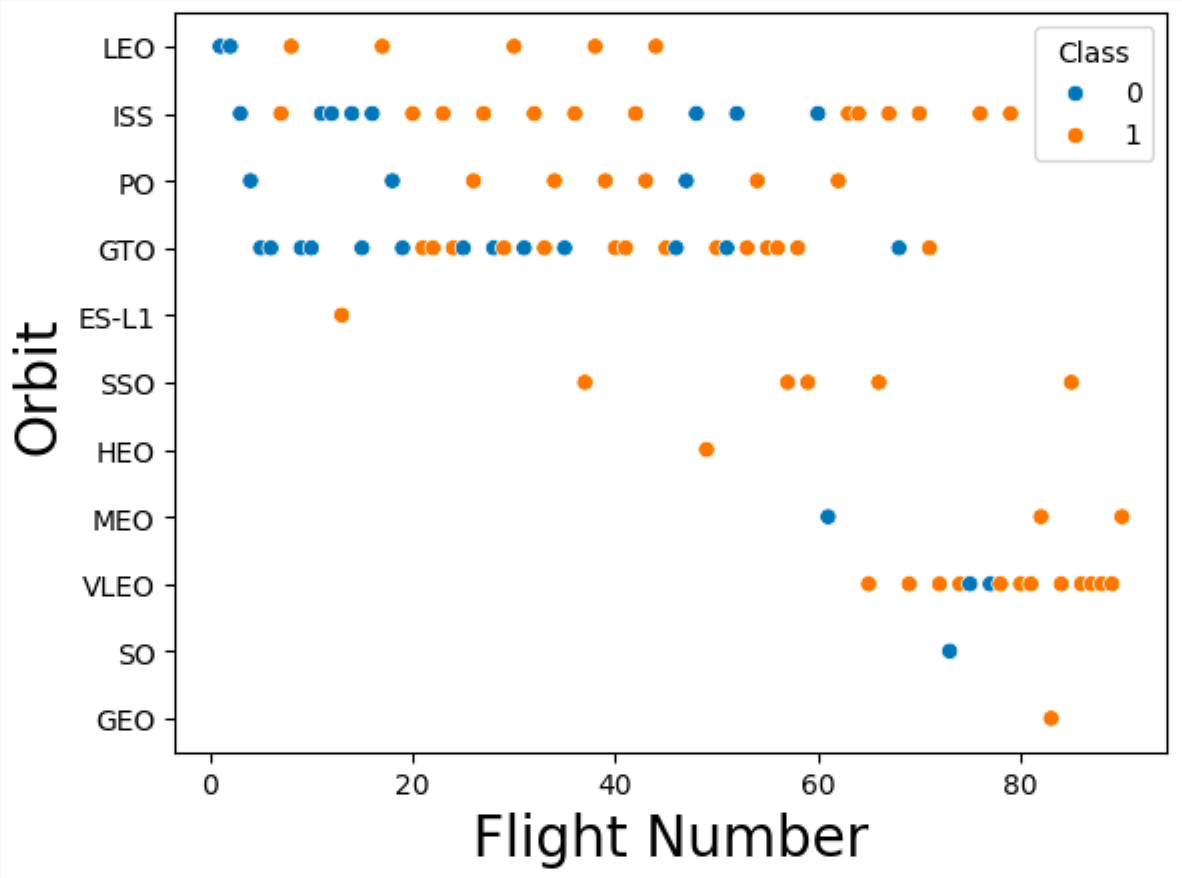
Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type



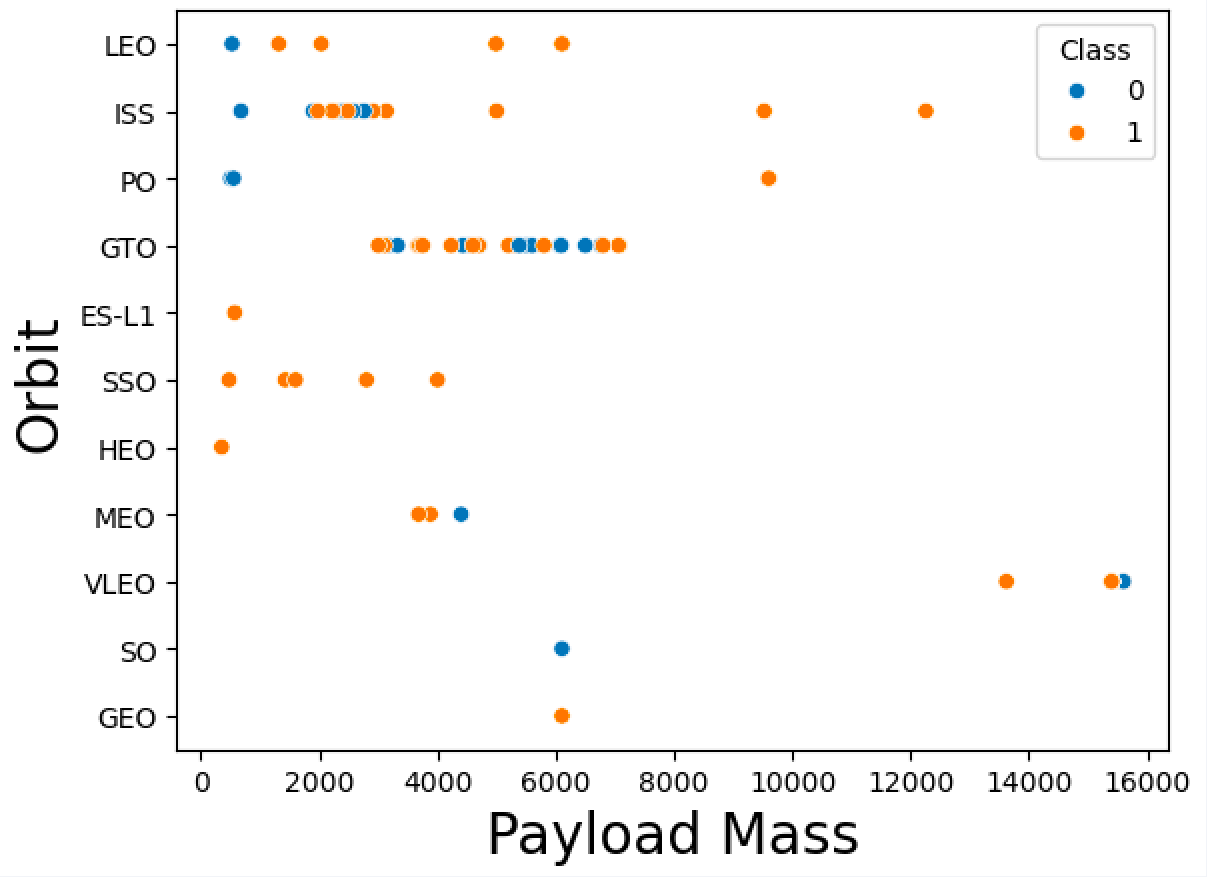
Orbits have the highest success rates are ES-L1, GEO, HEO and SSO.

Flight Number vs. Orbit Type



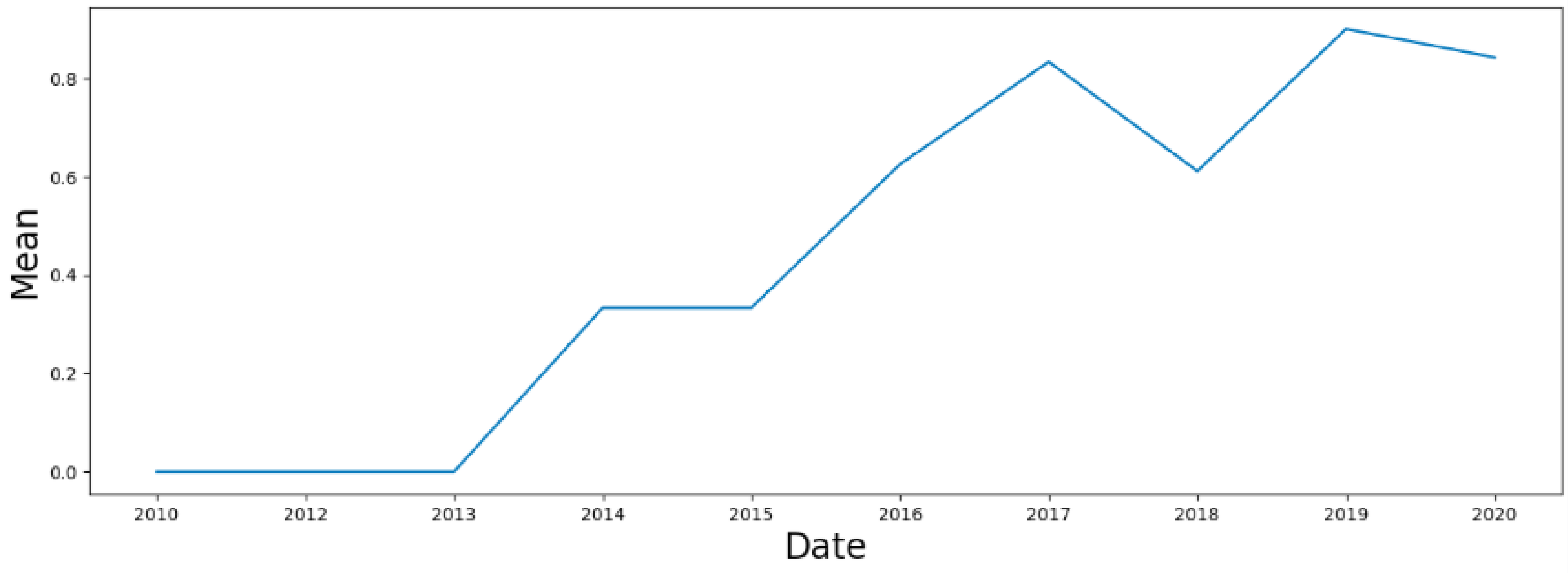
You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.
However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



You can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

```
[10]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE

 * sqlite:///my_data1.db
Done.
```

[10]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Using SELECT DISTINTC we can filter the results

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE '%CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success

Using SELECT, WHERE, LIKE and LIMIT we can filter the results.

Total Payload Mass

```
[29]: %sql SELECT SUM("PAYLOAD_MASS__KG_") FROM (SELECT * FROM SPACEXTABLE WHERE "Customer" LIKE "NASA (CRS)")

 * sqlite:///my_data1.db
Done.

[29]: SUM("PAYLOAD_MASS__KG_")
                        45596
```

We obtain the results using nested queries or subqueries. Additionally, we use the SQL SUM function.

Average Payload Mass by F9 v1.1

```
[31]: %sql SELECT AVG("PAYLOAD_MASS__KG_") FROM (SELECT * FROM SPACEXTABLE WHERE "Booster_Version" LIKE "%F9 v1.0%")

 * sqlite:///my_data1.db
Done.
```

[31]:

AVG("PAYLOAD_MASS__KG_")
340.4

We obtain the results using nested queries or subqueries. Additionally, we use the SQL AVG function.

First Successful Ground Landing Date

```
[34]: %sql SELECT MIN("Date") FROM (SELECT * FROM SPACEXTABLE WHERE "Landing_Outcome" LIKE "Success (ground pad)")

 * sqlite:///my_data1.db
Done.

[34]: MIN("Date")
      2015-12-22
```

We obtain the results using nested queries or subqueries. Additionally, we use the SQL MIN function.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
[36]: %sql SELECT "Booster_version", "Landing_Outcome", "PAYLOAD_MASS__KG_" FROM SPACEXTABLE WHERE ("Landing_Outcome

 * sqlite:///my_data1.db
Done.
```

[36]:

Booster_Version	Landing_Outcome	PAYLOAD_MASS__KG_
F9 FT B1032.1	Success (ground pad)	5300
F9 B4 B1040.1	Success (ground pad)	4990
F9 B4 B1043.1	Success (ground pad)	5000

We obtain the results using nested queries or subqueries. Additionally, we use the SQL BETWEEN and AND function.

Total Number of Successful and Failure Mission Outcomes

```
[63]: %sql SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTABLE WHERE "Mission_Outcome" == "Success"

 * sqlite:///my_data1.db
Done.
```

[63]:

Mission_Outcome	COUNT(*)
Success	98

```
[62]: %sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") FROM SPACEXTABLE WHERE "Mission_Outcome" != "Success"

 * sqlite:///my_data1.db
Done.
```

[62]:

Mission_Outcome	COUNT("Mission_Outcome")
Failure (in flight)	3

We obtain the results using nested queries or subqueries. Additionally, we use the SQL COUNT function and Operator.

Boosters Carried Maximum Payload

```
[81]: %sql SELECT "Booster_Version", "PAYLOAD_MASS__KG_" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" == (SELECT MAX('
 * sqlite:///my_data1.db
Done.
```

[81]:

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1060.3	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1056.4	15600
F9 B5 B1051.6	15600
F9 B5 B1051.4	15600
F9 B5 B1051.3	15600
F9 B5 B1049.7	15600
F9 B5 B1049.5	15600
F9 B5 B1049.4	15600
F9 B5 B1048.5	15600
F9 B5 B1048.4	15600

We obtain the results using nested queries or subqueries. Additionally, we use the SQL MAX, ORDER BY and DESC function and Operator.

2015 Launch Records

```
[33]: %%sql SELECT SUBSTR("Date", 0, 5) AS "Year", SUBSTR("Date", 6, 2) AS "Month", "Landing_Outcome", "Booster_Vers
      "Launch_Site" FROM SPACEXTABLE
      WHERE "Date" LIKE '2015%'

 * sqlite:///my_data1.db
Done.
```

[33]:

Year	Month	Landing_Outcome	Booster_Version	Launch_Site
2015	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015	02	Controlled (ocean)	F9 v1.1 B1013	CCAFS LC-40
2015	03	No attempt	F9 v1.1 B1014	CCAFS LC-40
2015	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
2015	04	No attempt	F9 v1.1 B1016	CCAFS LC-40
2015	06	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40
2015	12	Success (ground pad)	F9 FT B1019	CCAFS LC-40

We obtain the results using SQL SUBSTR, WHERE and LIKE function and Operator.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
[24]: %%sql SELECT "Landing_Outcome", COUNT(*) FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY COUNT (*) DESC

 * sqlite:///my_data1.db
Done.
```

[24]:

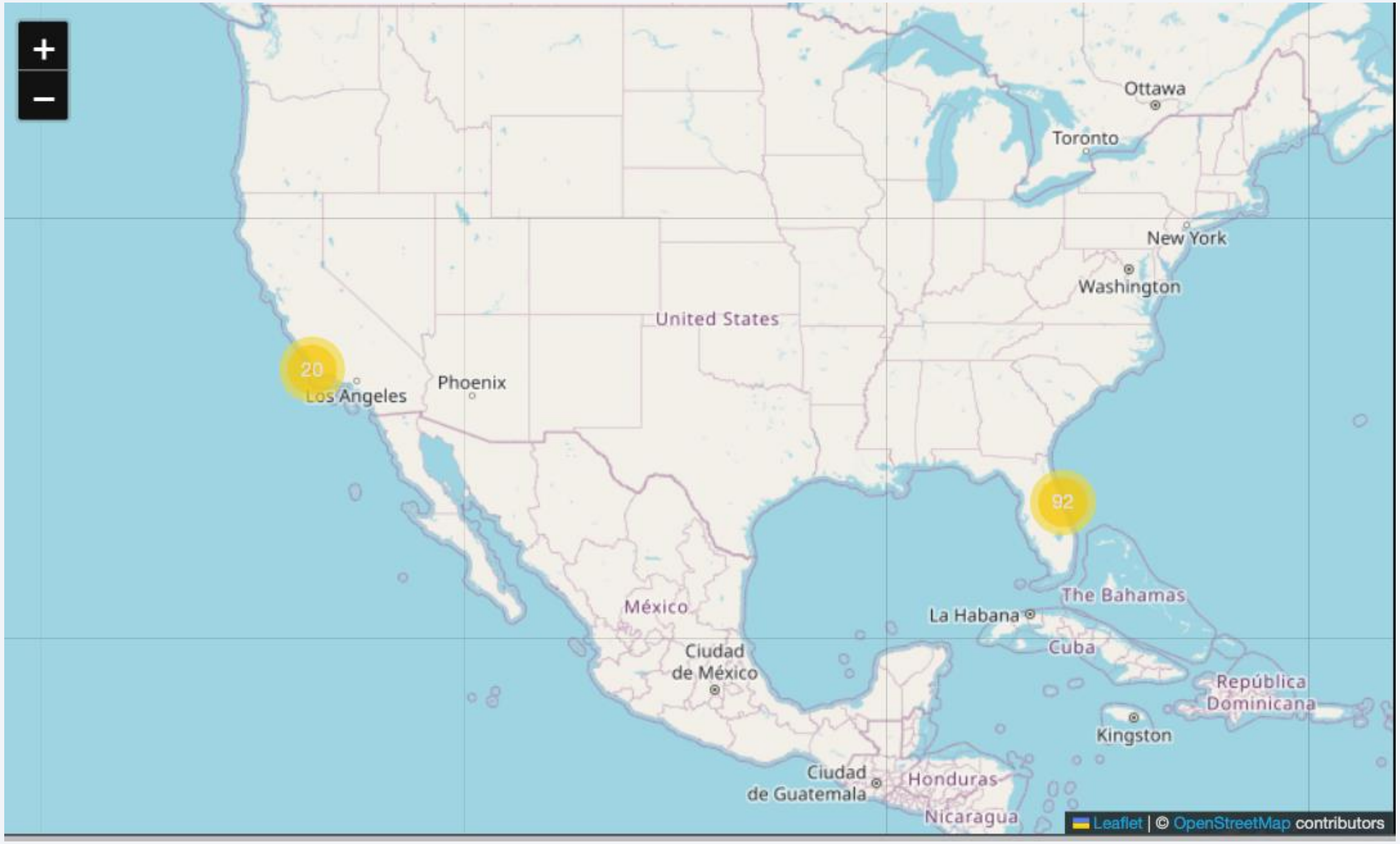
Landing_Outcome	COUNT(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

We obtain the results using SQL COUNT, WHERE, BETWEEN, AND, GROUP BY, ORDER BY and DESC function.

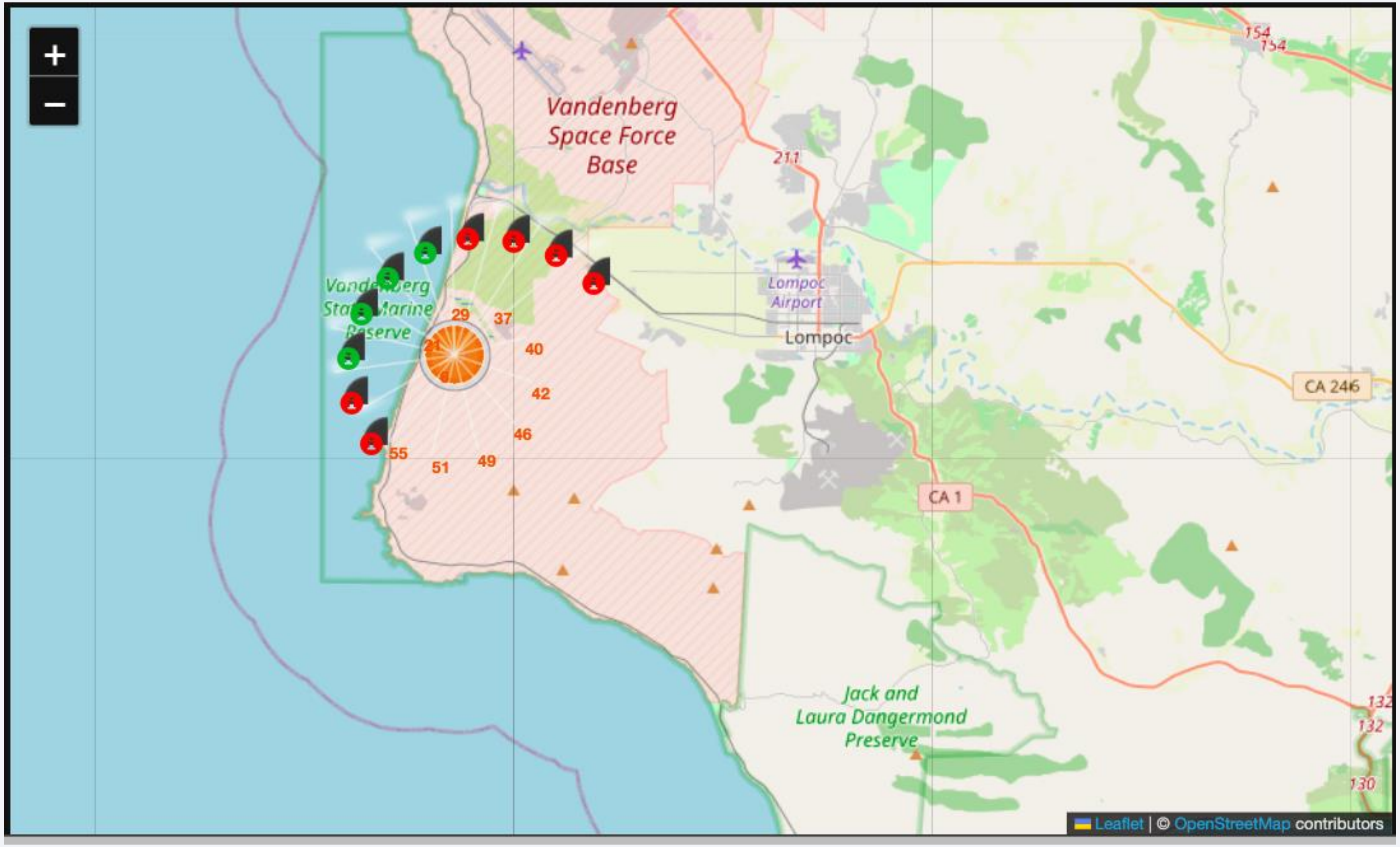
Section 3

Launch Sites Proximities Analysis

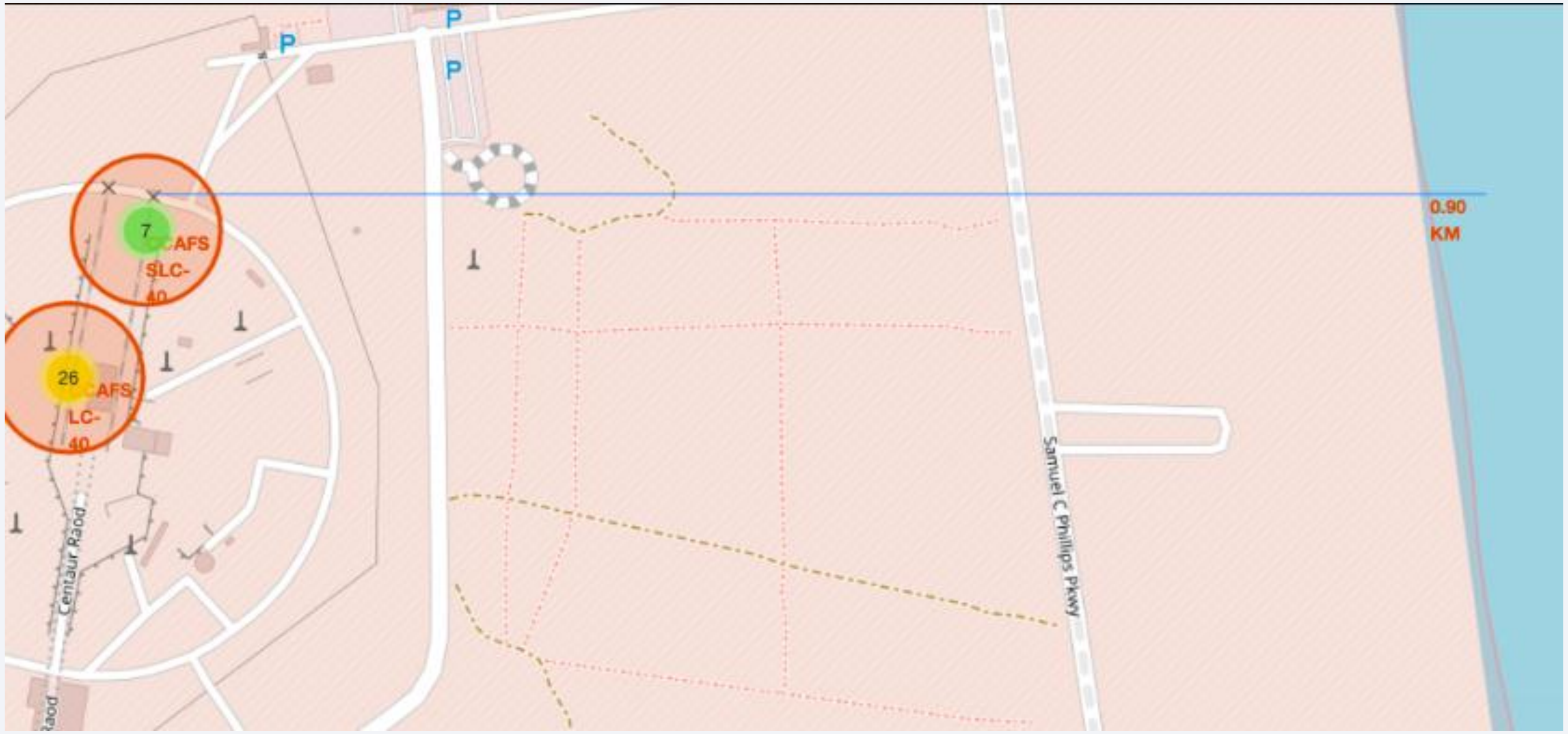
Folium map with all launch sites



Folium map with color labeled launch outcomes

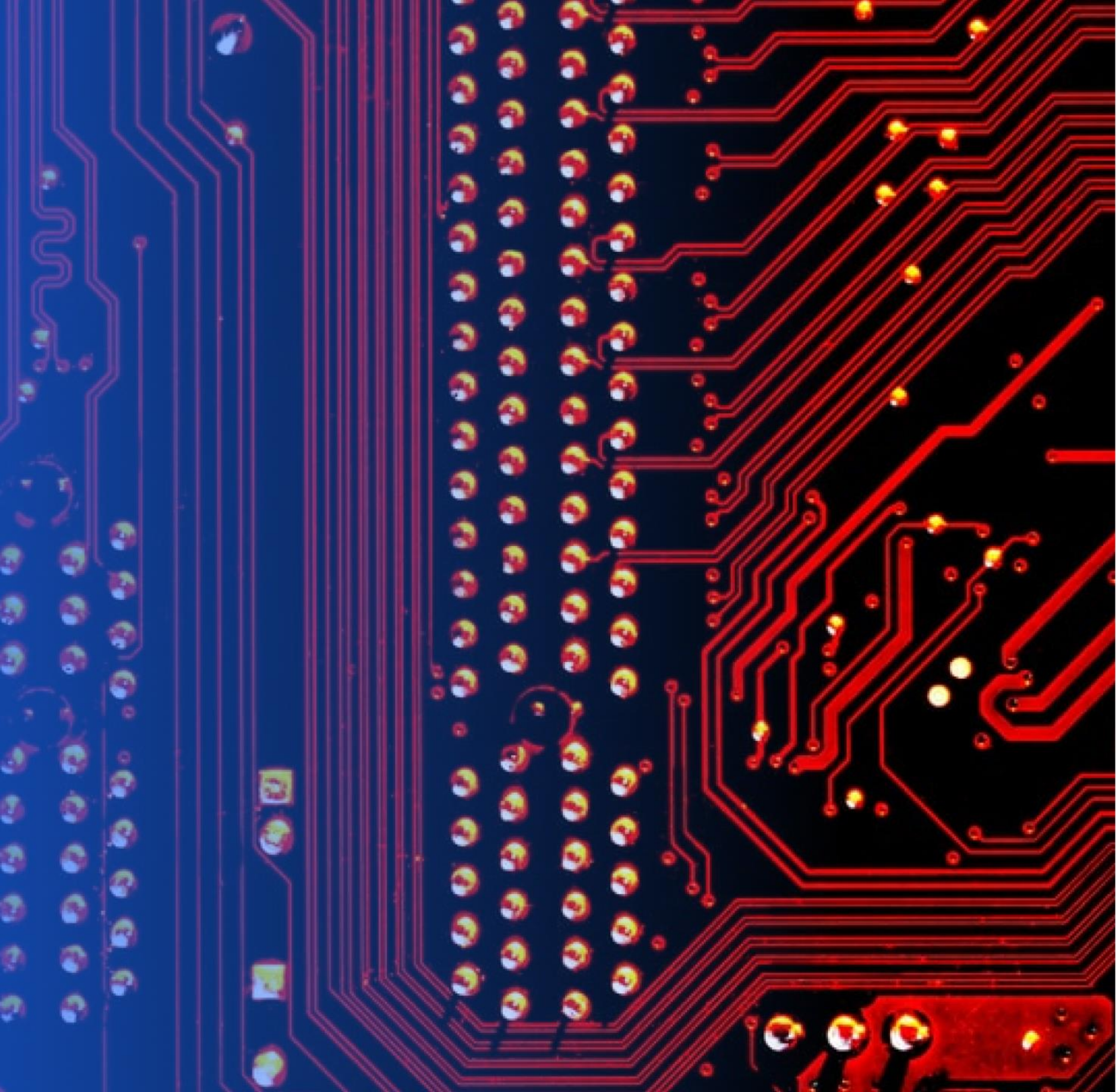


Folium map with launch site proximities

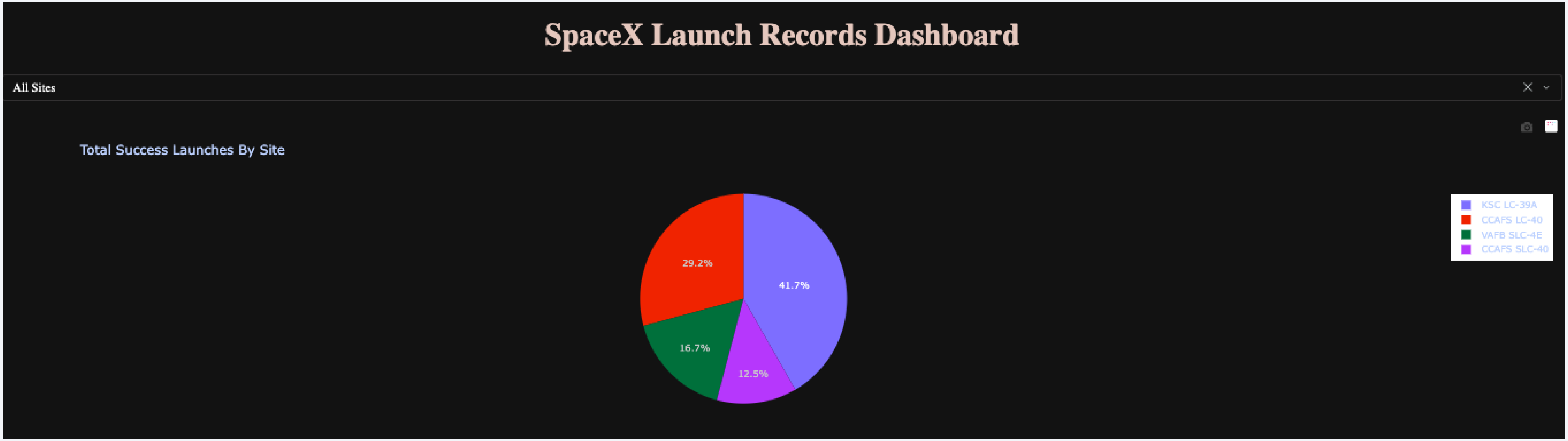


Section 4

Build a Dashboard with Plotly Dash



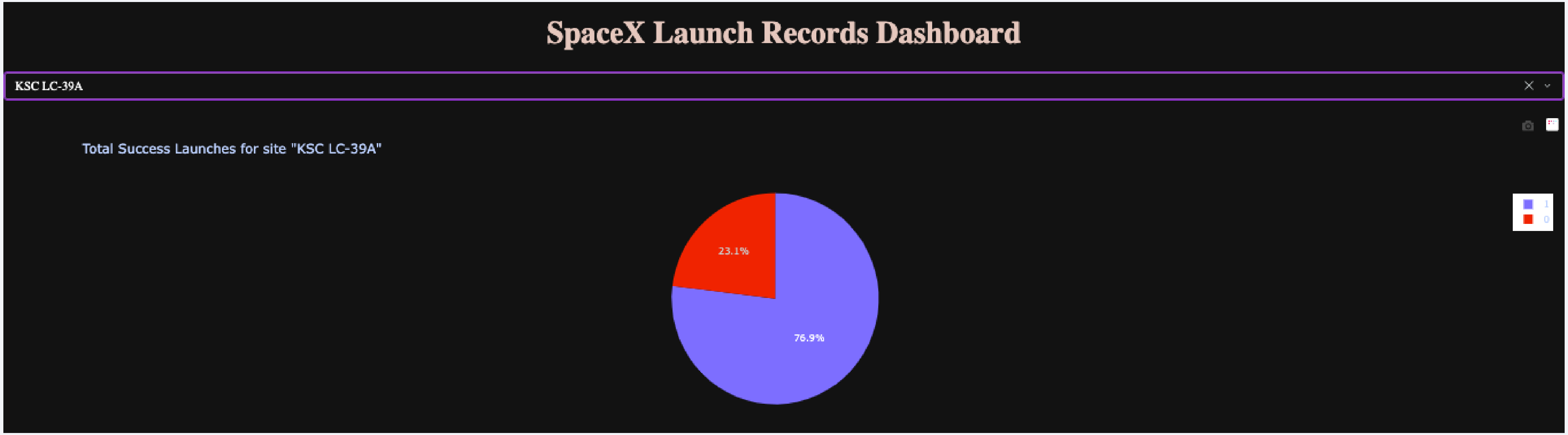
Launch success count for all sites



We conducted this lab to analyze and answer the following:

- Which site has the largest successful launches? KSC LC-39A with10 launches
- Which site has the highest launch success rate? KSC LC-39A with 41,70%

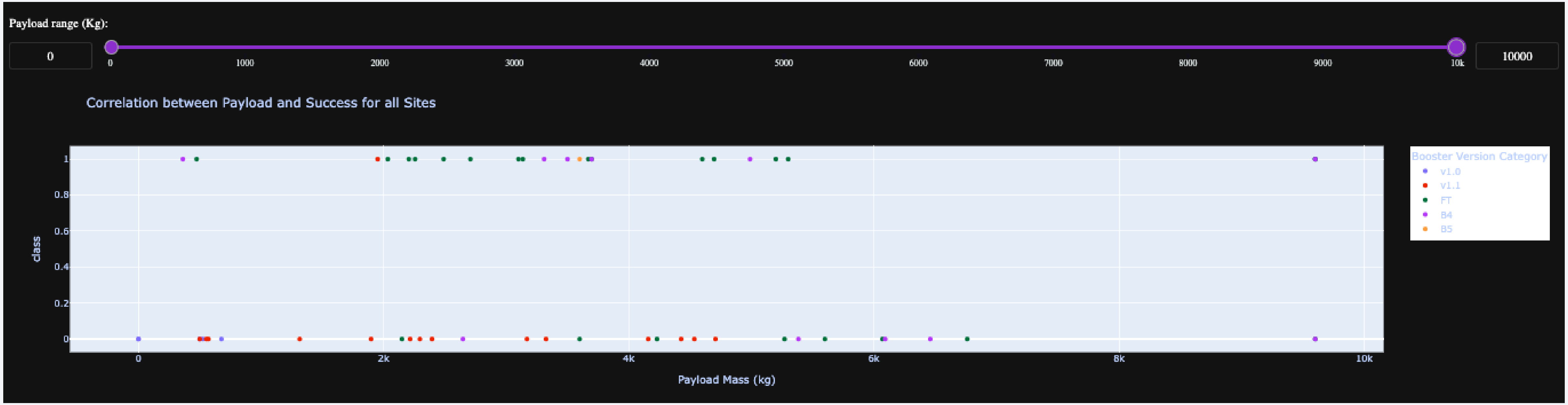
Launch success ratio for KSC LC-39A site



We conducted this lab to analyze and answer the following:

- Which KSC LC-39A site has successful launches? 10 launches
- Which KSC LC-39A site has launch success rate? 76,90% success rate

Payload vs. Launch Outcome for all sites



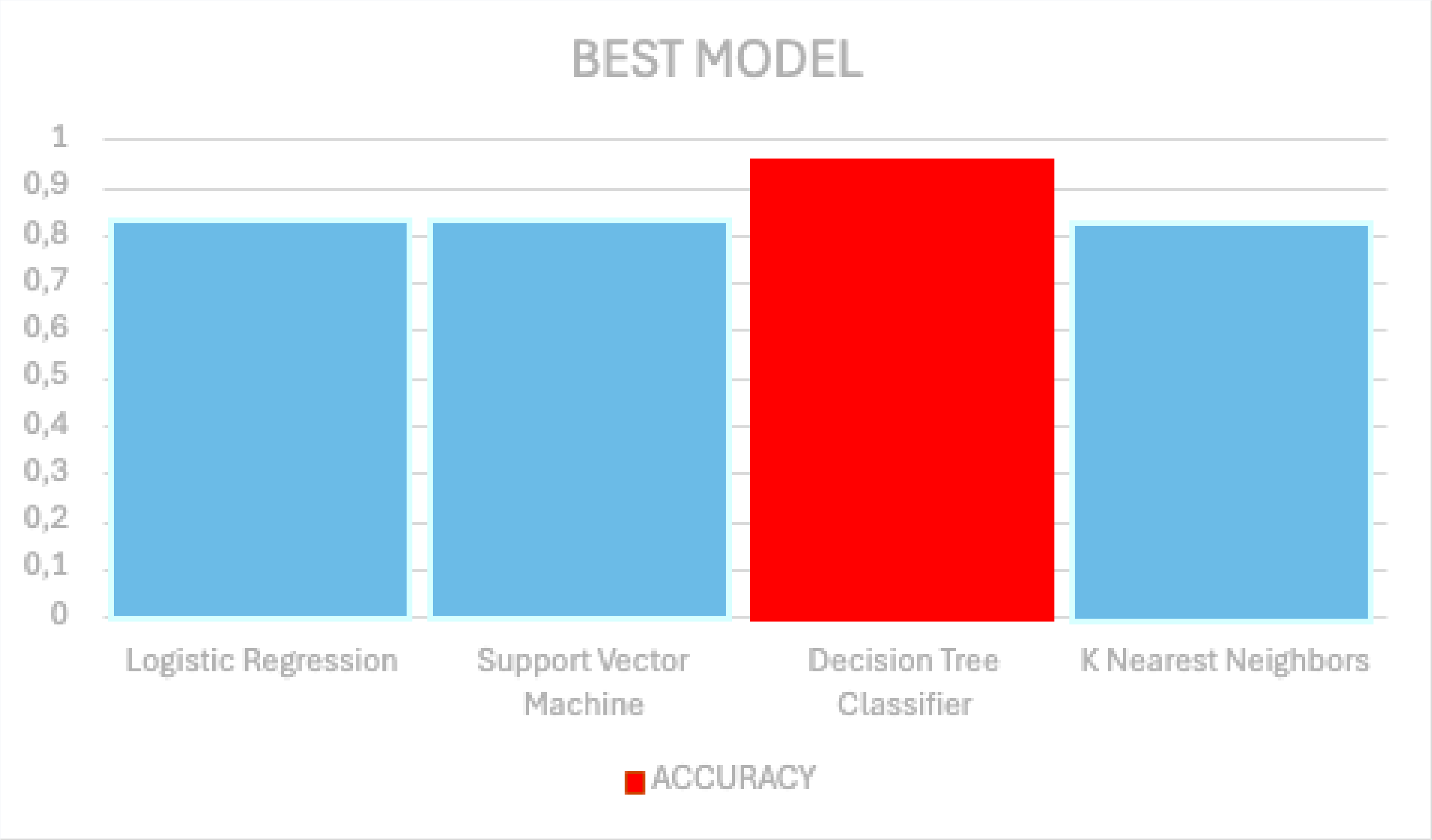
We conducted this lab to analyze and answer the following:

- Which payload range(s) has the highest launch success rate? From 2K to 4K
- Which payload range(s) has the lowest launch success rate? From 9K to 10K
- Which F9 Booster version (v1.0, v1.1, FT, B4, B5, etc.) has the highest launch success rate? FT

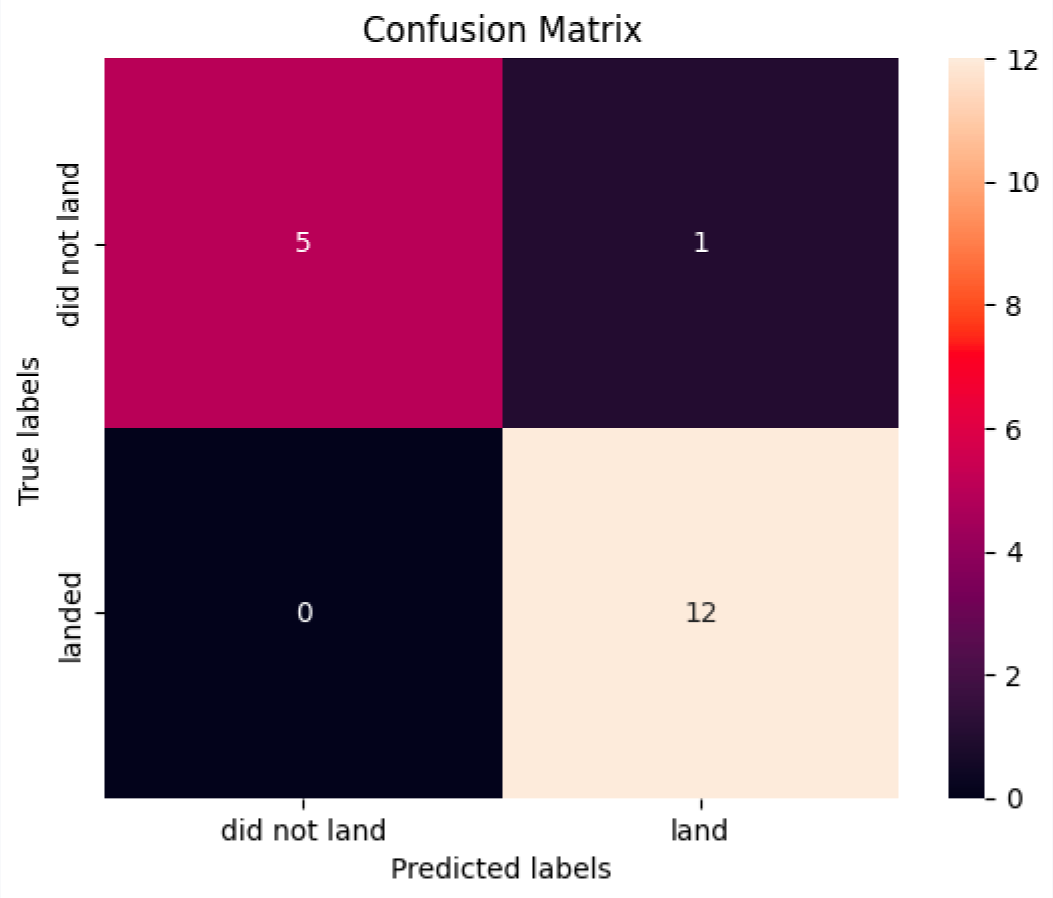
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



The method performs best is Decision Tree Classifier with Accuracy: 0.9444444444444444 and False Positive = 1

Conclusions

The method performs best is Decision Tree Classifier with Accuracy: 0.9444444444444444 and False Positive = 1

Appendix

- https://github.com/controlparen/Proyect

Thank you!